

CS-UY 2214 — Recitation 5

Introduction

Complete the following exercises. Unless otherwise specified, put your answers in a plain text file named `recitation5.txt`. Number your solution to each question. When you finish, submit your file on Gradescope. Then, in order to receive credit, you must ask your TA to check your work. Your work should be completed and checked during the recitation session.

You may consult the E20 manual, which is available on Brightspace.

Problems

1. Consider the following complete E20 program.

```
movi $1, foo           # ram[0]
lw $2, foo($0)         # ram[1]
halt                   # ram[2]
foo:
    .fill 42            # ram[3]
bar:
```

- (a) What is the value of the label `foo`? What is the value of the label `bar`?
- (b) Assume that the initial value of all registers is 0. What will be the final value of the program counter, `$1`, and `$2`?
- (c) Convert this program into E20 machine code.

2. Consider the following complete E20 program.

```
movi $1, foo           # ram[0]
addi $1, $1, 1         # ram[1]
jr $1                  # ram[2]
halt                   # ram[3]
foo:
    movi $2, 2          # ram[4]
    movi $3, 3          # ram[5]
    halt               # ram[6]
```

- (a) Assume that the initial value of all registers is 0. What will be the final value of the program counter, `$1`, `$2`, and `$3`?
- (b) Convert this program into E20 machine code.

3. Consider the following complete E20 program.

```

    movi $1, 7
    add $2, $1, $1
    sw $1, foobar($0)
    sw $2, foobar($1)
    halt

foobar:
    .fill 600
    .fill 700

```

What memory addresses will be modified by this program, and what will their final value be?

4. Consider the following complete E20 program.

```

main:
    movi $1, 0          # ram[0]
    movi $7, 0          # ram[1]
repeat:
    lw $2, array($1)    # ram[2]
    jeq $2, $0, done    # ram[3]
    slt $5, $7, $2      # ram[4]
    jeq $5, $0, next    # ram[5]
    add $7, $0, $2      # ram[6]
next:
    addi $1, $1, 1      # ram[7]
    j repeat            # ram[8]
done:
    halt                # ram[9]
array:
    .fill 53            # ram[10]
    .fill 22            # ram[11]
    .fill 94            # ram[12]
    .fill 2             # ram[13]
    .fill 19            # ram[14]
    .fill 0             # ram[15]

```

Assume that the initial value of all registers is 0. What will be the final value of \$7? In English, what does this program do? Discuss how it works.

5. Consider the following complete E20 program.

```

foo:
    movi $1, foo        # ram[0]
    lw $2, foo($0)      # ram[1]
    halt                # ram[2]

```

Assume that the initial value of all registers is 0. What will be the final value of the program counter, \$1, and \$2?

Hint: compare this question to question 1.

6. Consider that we have two unknown values in labeled memory locations:

```

first: .fill somenumber
second: .fill anothernumber

```

Write a program that will swap the values of these cells. That is, when your program ends, the value *another number* will be in the cell identified by the label **first**; and the value *some number* will be in the cell identified by the label **second**.